

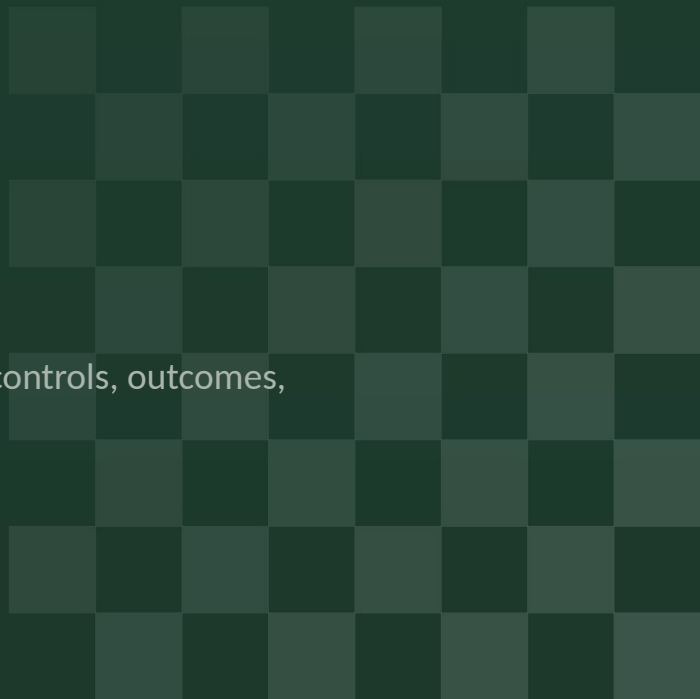


DATA VISUALISATION • FINAL PROJECT

Chess at Scale

A Lichess Analysis of 6.24 Million Games

Interactive dashboard and notebook analysis — ratings, openings, time controls, outcomes, and play patterns





Project Overview

6.24M

finished Lichess games analysed

1741

average player Elo rating

49.67%

of games won by White

3.83%

of games end in a draw

Sicilian

most common opening family

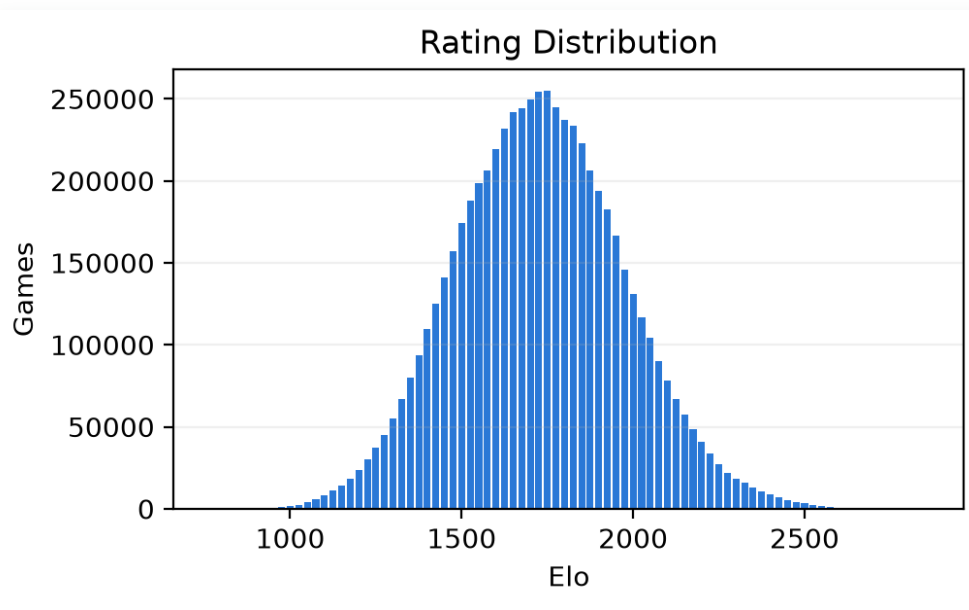
12

analysis questions answered



Rating Distribution

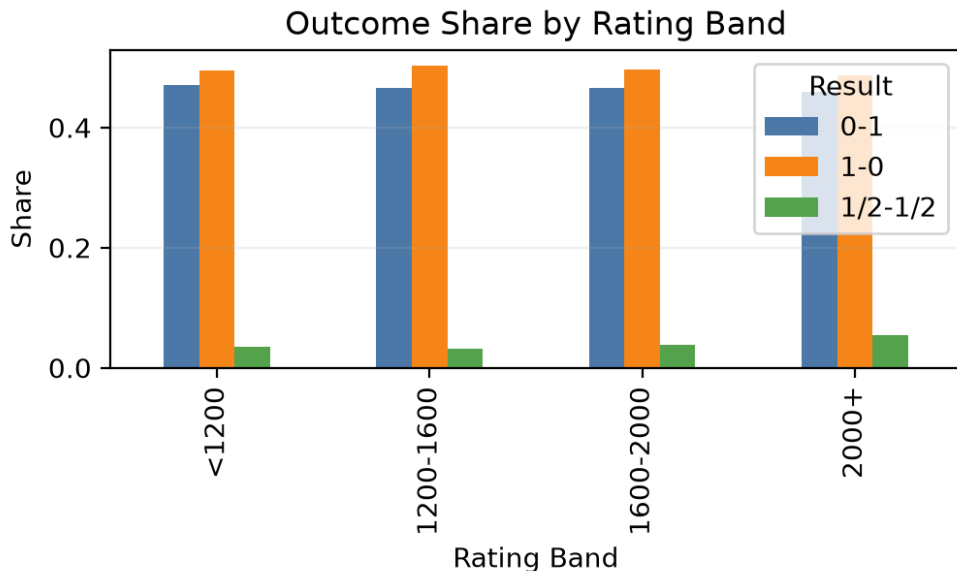
- Ratings are concentrated in the 1200–2200 range.
- The distribution shows strong participation from intermediate and advanced players.
- The dashboard uses this view to compare player skill bands.





OUTCOMES

Outcome Shares by Rating Band

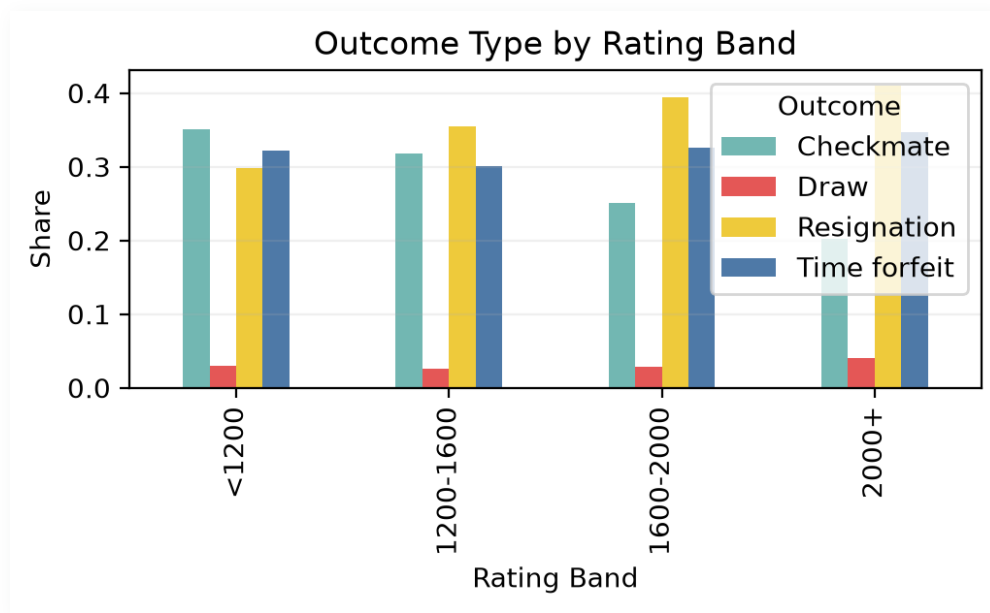


- White wins are slightly more common overall.
- Higher-rated players show more decisive outcomes and fewer draws.
- Stronger players convert their advantage more often.



How Games End

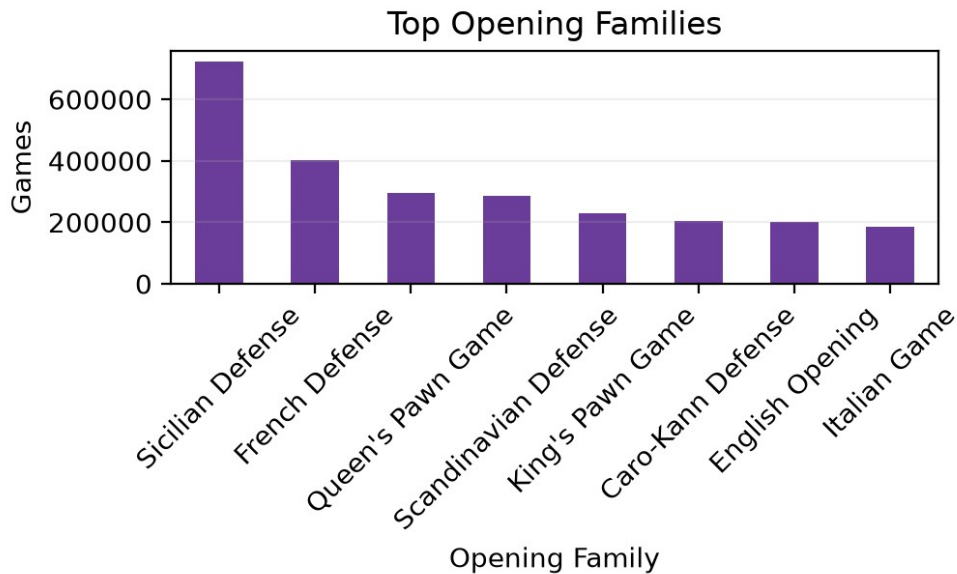
- Resignation and checkmate dominate the result types.
- Time forfeit is common in lower-rated groups.
- Lower-rated games are often decided by practical errors rather than pure tactical endings.





OPENINGS

Opening Preferences

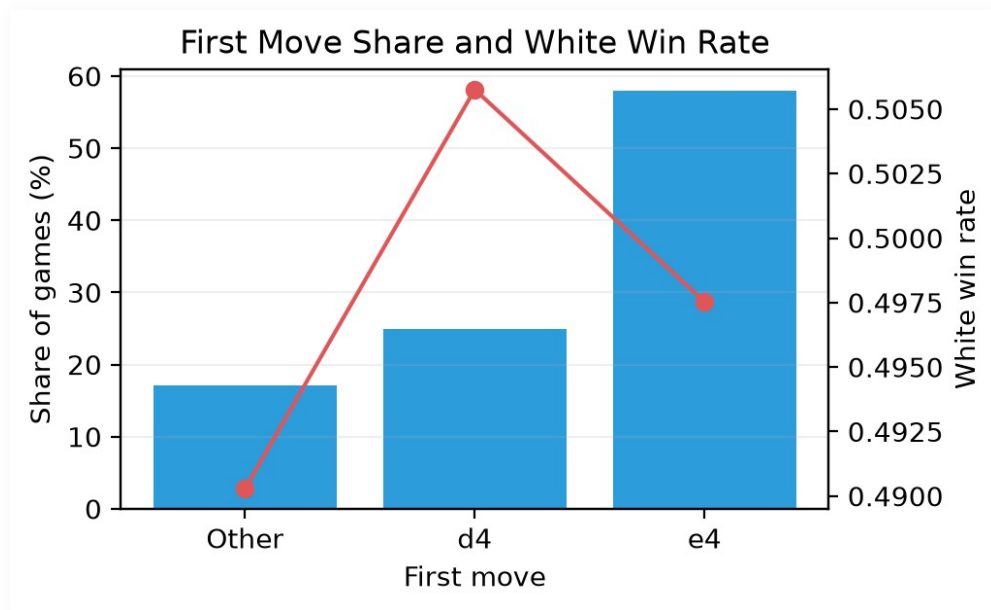


- The Sicilian Defense is the most prominent opening in the dataset.
- Opening choice changes by rating band, suggesting different strategic preferences.
- Openings are a strong lens for comparing player styles.



First-Move Choices

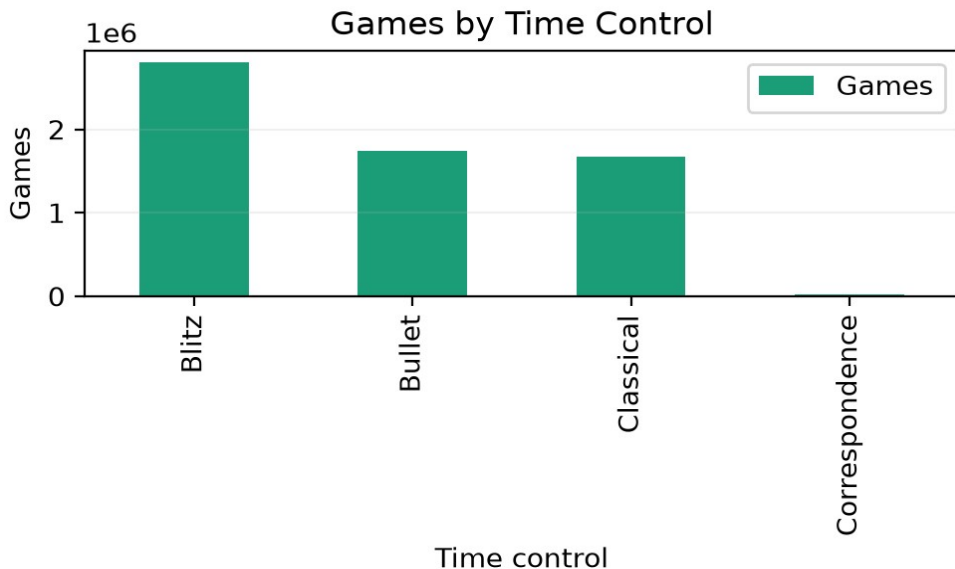
- 1.e4 is the most common first move — about 58% of games.
- 1.d4 is next most common and shows a slightly higher White win rate.
- This makes the opening phase a key part of the interactive dashboard.





TIME CONTROLS

Time Controls

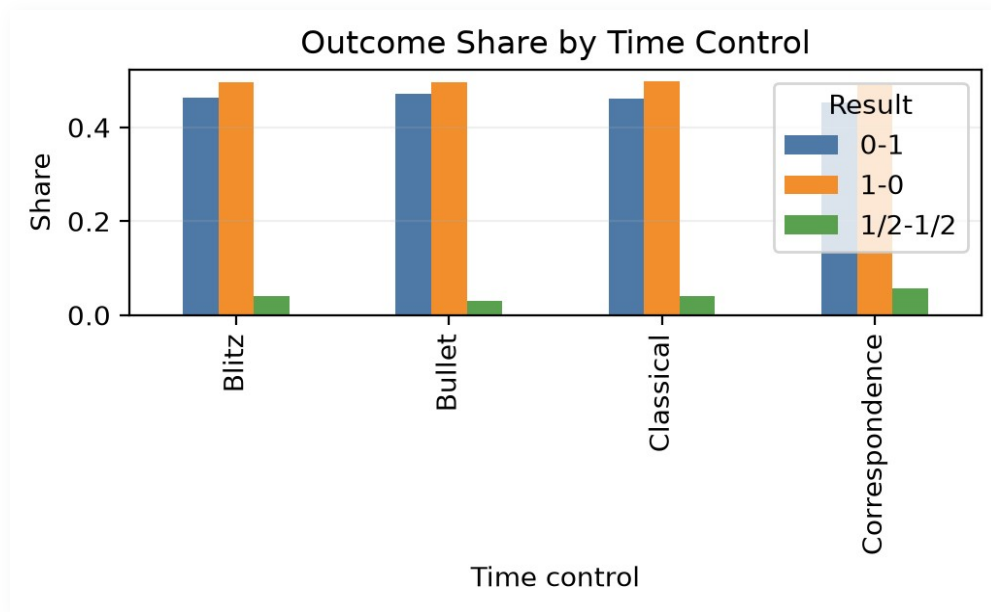


- Blitz is the most common format, followed by Bullet and Classical.
- Correspondence is far less frequent but still part of the dataset.
- Rapid online chess dominates the sample.



Fast Games, Decisive Results

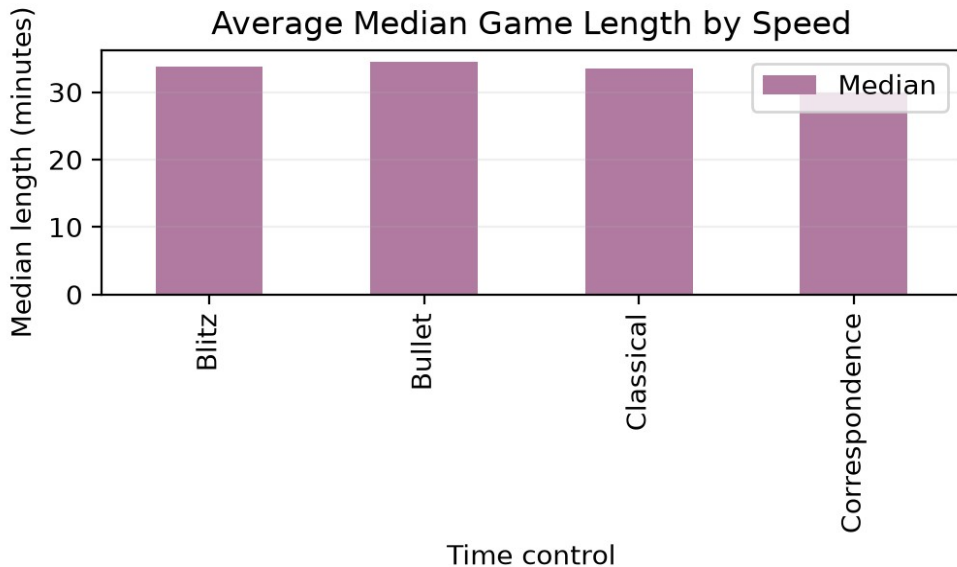
- Faster formats produce a slightly higher share of decisive outcomes.
- Bullet and Blitz show less draw tendency than slower formats.
- Pace clearly shapes the style of play.





PACE

Game Length and Pace

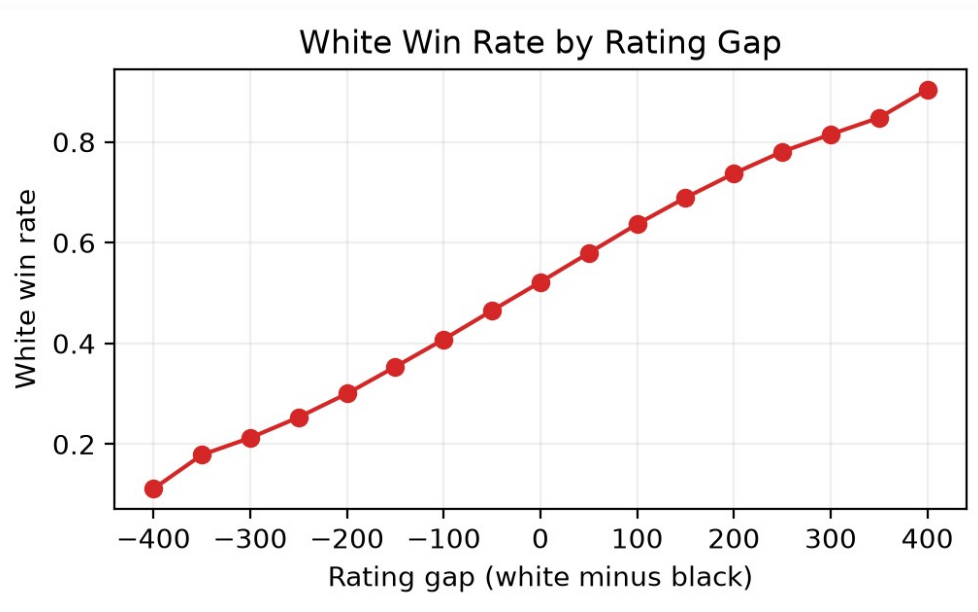


- Median game length rises as the format becomes slower.
- Higher-rated players tend to play longer games within the same speed category.
- Skill and time control together shape game structure.



Rating Gap and Win Probability

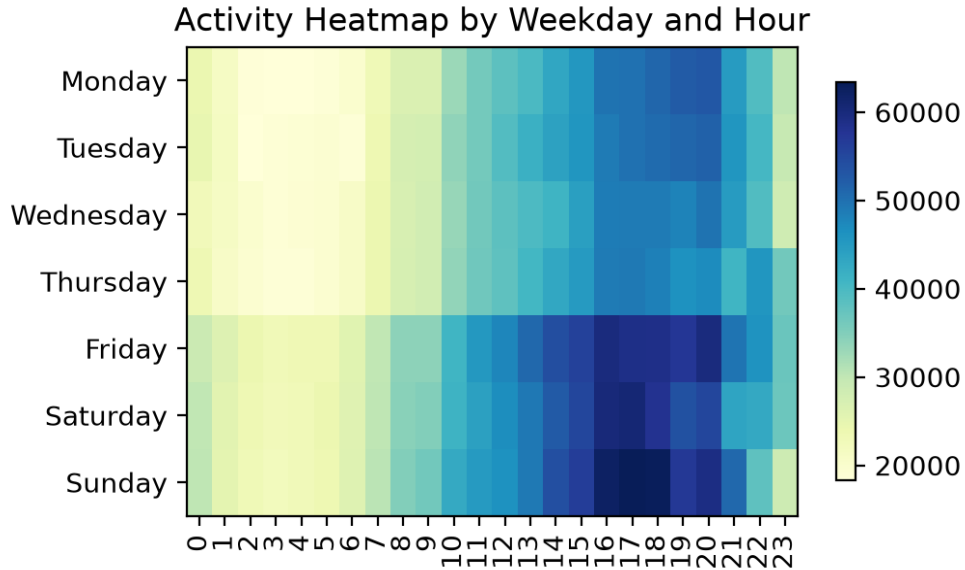
- As the rating advantage grows, the stronger side wins more often.
- Rating is a strong predictor of game outcome.
- The dashboard makes this relationship easy to explore interactively.





ACTIVITY

When Are Games Played?



- The heatmap shows clear weekly and hourly activity patterns.
- Online chess activity is concentrated in the evenings and at weekends.
- This connects gameplay behaviour with audience and engagement patterns.

CONCLUSIONS

Key Insights

1

Outcomes are shaped by fundamentals

Rating, opening choice, time control, and game pace all leave a measurable mark on how games end.

2

Rating is the strongest predictor

The win-probability curve rises steadily with rating advantage — skill converts to results.

3

Pace changes the game

Faster formats are more decisive; slower formats produce longer, more balanced games.

4

Data becomes a story

The interactive dashboard turns 6.24M games into explorable, understandable insights.



Explore the Project

All code, data preparation, and the live dashboard are public.

GITHUB REPOSITORY

github.com/shreedevale-create/DV_Project

LIVE STREAMLIT DASHBOARD

[dvproject...streamlit.app](https://dvproject.streamlit.app)